Effects of fermented corn cob on growth performance, digestive enzyme, immune response, and gene expression of nile tilapia (*Oreochromis niloticus*) raised in biofloc system

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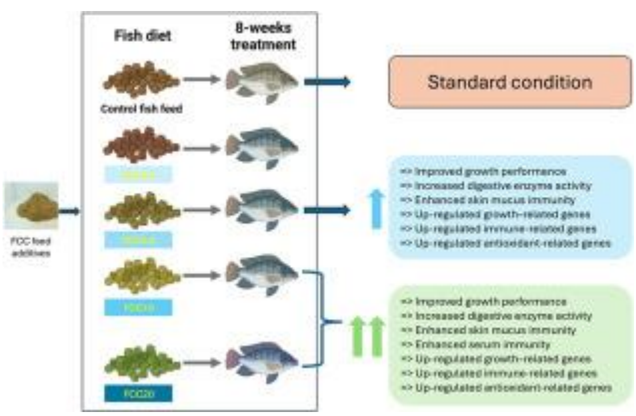
Highlights

- FCC 10–20 g kg^{-1} improved growth and feed efficiency in Nile tilapia under biofloc conditions.
- FCC at 10–20 g kg^{-1} enhanced digestive enzyme activities (amylase, protease, trypsin, and lipase).
- FCC at 10–20 g kg^{-1} enhanced lysozyme, peroxidase, and complement activity in skin mucus and serum.
- FCC at 10–20 g kg^{-1} upregulated genes related to growth, immunity-, and antioxidants in head kidney and intestinal tissues.

Abstract

Utilizing agricultural by-products through fermentation presents a sustainable strategy to enhance the nutritional value of aquaculture feeds. This study aimed to evaluate the effects of different dietary levels of fermented corn cob (FCC) on growth performance, digestive enzyme activities, immune responses, and gene expression in Nile tilapia cultured in a biofloc environment. Three hundred Nile tilapia fingerlings (average of 37.58 ± 0.12 g) were randomly allocated to five dietary treatments (0, 2.5, 5, 10, and 20 g kg^{-1} FCC) with three replicates per treatment. Fish were reared in 150-L aquaria under biofloc conditions for eight weeks. Growth performance, digestive enzyme activities, innate immune parameters in skin mucus and serum, as well as gene expressions were measured. The results indicated that fish fed diets containing 10–20 g kg^{-1} FCC exhibited significantly greater final weight, weight gain, and specific growth rate compared to the control group ($P < 0.05$), along with improved feed conversion ratios. Digestive enzyme activities were significantly enhanced in fish receiving FCC10 and FCC20 diets. Similarly, innate immune responses, including lysozyme, peroxidase, and complement activity, were significantly upregulated in both skin mucus and serum of FCC-fed fish. At the molecular level, the expression of growth-related genes (ghrelin, galanin, *EF-α,* and *NPY-α*), immune-related genes (*il-1β, MHC II-α, TNF-α,* and *NFκB*), antioxidant-related genes (*GPX, hsp70,* and *nrf2*) was significantly upregulated in fish fed FCC10 and FCC20 diets. Overall, dietary supplementation with 10–20 g kg^{-1} FCC under biofloc conditions significantly enhanced growth performance, digestive function, innate immunity, and gene expression profiles in Nile tilapia, supporting the potential application of FCC as a sustainable functional feed additive to promote health and productivity in aquaculture.

Graphical abstract



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Introduction

Nile tilapia (*Oreochromis niloticus*) is one of the most widely farmed freshwater species globally due to its rapid growth, high adaptability to diverse environmental conditions, and strong resistance to diseases [1]. It holds significant economic importance in the aquaculture industry and serves as an affordable and accessible source of high-quality dietary protein, particularly in developing countries where it contributes substantially to food security and nutritional health [2,3]. Its production steadily increased from 4.53 million metric tons in 2018 to 5.3 million metric tons in 2022 [4]. Furthermore, the global tilapia market is projected to grow at a compound annual growth rate (CAGR) of 3.9 %, reaching an estimated value of US$10.2 billion by 2028 [5]. This consistent increase in demand has led to the intensification of tilapia farming worldwide [6]. Despite its global importance, tilapia farming faces several challenges that hinder profitability [7]. Feed accounts for up to 70 % of total production expenses in aquaculture, largely influenced by dietary protein levels and the sources of feed ingredients, whether plant- or animal-based [8]. One of the most critical challenges is the high cost and limited availability of fishmeal (FM) and sustainable, high-yielding plant protein sources to replace FM [9,10]. Additionally, rising global feed prices, driven by competition for plant proteins and grains in bioenergy production, have intensified the need to identify unconventional feed resources [11,12].

Among agricultural by-products, maize (*Zea mays*), the world's second most cultivated crop [13] generates significant quantities of agricultural waste in the form of maize cobs [14], with approximately 164 millions tons of maize by-products produced annually [15]. It is estimated that around 180 kg of cobs are generated per ton of maize harvested [16,17]. While often discarded, maize cobs are rich in bioactive compounds such as phenolic acids and flavonoids, known for their antibacterial, anti-inflammatory, antioxidant, anticancer, antidiabetic, antithrombotic, and neuroprotective properties [18]. Despite these attributes, the high fiber content and low protein value of maize cobs limit their direct use in aquafeeds, particularly for monogastric species such as fish [19]. However, recent advancements in fermentation technologies, notably solid-state fermentation (SSF) using probiotics such as yeast, fungi, and bacteria, have demonstrated the potential to enhance the nutritional quality of agricultural by-products [20]. SSF can increase protein levels, reduce fiber, lipid, and ash content, and eliminate anti-nutritional factors like phytic acid [21,22], thereby improving nutrient digestibility and making maize cobs a viable feed component [23].

In parallel, biofloc technology (BFT) has gained attention an environmentally sustainable approach to managing water quality and nutrient recycling in aquaculture systems [24]. BFT enhances water quality by converting toxic nitrogen compounds, such as ammonia and nitrite, into microbial biomass, thus reducing water pollution [25]. This system relies on organic carbon source supplementation to stimulate heterotrophic bacterial growth, which in turn assimilates inorganic nitrogen to produce microbial protein [26]. This microbial biomass serve as a natural feed source for aquatic species, improving their nutritional intake and reducing reliance on commercial feeds [27]. Additionally, BFT minimizes water exchange requirements, promoting environmentally friendly and resource-efficient aquaculture practices [28,29]. The integration of BFT with functional feed additives offers a promising strategy to address critical challenges in aquaculture, including feed costs, water management, and environmental sustainability [24,30]. We hypothesize that incorporating fermented maize cob into biofloc systems could offer dual benefits: enhancing the nutritional composition of microbial flocs and leveraging the functional properties of maize cob to promote fish health and performance. Although this approach appears promising, no prior studies have evaluated the effects of fermented maize cob on Nile tilapia (*O. niloticus*), one of the most widely cultured aquaculture species globally. Therefore, this study investigates the effects of different inclusion levels of fermented maize cob on growth performance, digestive enzyme activity, immune responses, and gene expression in Nile tilapia reared in a biofloc system.

Sample preparation

Corn cobs were collected from the experimental farm at the Faculty of Agriculture, Chiang Mai University, Thailand. After collection, the cobs were oven-dried at 60 °C for 48 h, ground with a hammer mill, and sieved through a 100-mesh screen to achieve uniform particle size. The processed corn cob powder was stored at 4 °C until further use. The proximate composition of the corn cobs, determined on a dry weight basis, was as follows: cellulose 45.78 %, hemicellulose 36.25 %, lignin 5.86 %, ash

Growth performance

The initial weight (IW) of Nile tilapia was similar across all dietary treatments, ranging from 37.52 to 37.70 g (Table 5). However, significant differences were observed in final weight (FW), weight gain (WG), specific growth rate (SGR), and feed conversion ratio (FCR) among the different inclusion levels of fermented corn cob (FCC). Notably, tilapia fed FCC10 and FCC20 had significantly higher FW compared to the control group (FCC0) and lower FCC levels (FCC2.5 and FCC5). Similarly, WG was

Discussion

This study demonstrates that supplementing Nile tilapia (*Oreochromis niloticus*) diets with fermented corn cob (FCC) enhances growth performance, immune responses, and growth-, immune-, and antioxidant-related gene expressions. Notably, fish receiving 10 and 20 g kg^{-1} FCC showed significant growth improvements over the control group. While this is the first investigation into FCC's effects on fish, similar benefits have been observed in other species, such as common carp (*Cyprinus carpio*) fed

Conclusion

This study demonstrates that dietary supplementation with fermented corn cob (FCC) in a biofloc system significantly enhances growth performance, digestive enzyme activity, and immune responses in Nile tilapia. Optimal inclusion levels (10–20 g kg^{-1}) improved both mucosal and systemic immunity, as evidenced by elevated lysozyme, peroxidase, and complement activity, along with the upregulation of immune- and antioxidant-related genes. These findings highlight FCC as a promising functional feed

CRediT authorship contribution statement

Hien Van Doan: Conceptualization, Methodology, Supervision, Writing – review & editing, Project administration, Funding acquisition. **Supriya Wannavijit:** Formal analysis. **Khambou Tayyamath:** Investigation, Data Collection. **Tran Thi Diem Quynh:** Data Collection. **Punika Ninyamasiri:** Data Collection. **Nguyen Vu Linh:** Data Collection. **Sutee Wongmaneeprateep:** Writing – review & editing. **Channarong Rodkhum:** Writing – review & editing. **Phisit Seesuriyachan:** Resources. **Yuthana Phimolsiripol:** Writing –

Declaration of Competing interest

The authors declare no conflict of interest.

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